

Crossed-Product Entropy Does Not Supply an Omitted Finite-Center Calibration

Live review note of June 20, 2026

Abstract

Gravitational crossed-product constructions define entropy for a specified Type II or semifinite von Neumann algebra together with its trace or normalization convention. This note isolates a finite-central boundary. If $\mathcal{M}$ is replaced by

$$\mathcal{B}_n = \mathcal{M} \bar{\otimes} \ell^\infty(\{1, \dots, n\}),$$

then the modular crossed product carries the finite center to the Type II core:

$$\mathcal{B}_n \rtimes_{\sigma^{\Omega_r}} \mathbb{R} \cong (\mathcal{M} \rtimes_{\sigma^\omega} \mathbb{R}) \bar{\otimes} \ell^\infty(\{1, \dots, n\}).$$

Trace-normalized Type II entropy fixes the finite-center entropy contribution only when the supplied trace or absolute calibration separates the central weights relevant to the claimed state family. Therefore Witten-style crossed-product entropy is a positive entropy construction for the supplied algebra and trace; it is not, by itself, a principle that reconstructs an omitted finite central summand, finite central trace weights, or an absolute entropy-density coefficient. The positive exit is equally concrete: atomic trace calibration on the retained finite central summands recovers the finite-center entropy term up to the intended common trace convention.

Status. This is non-frozen external-review metadata for Team 1. It is intended as a small claim that a crossed-product, Type II, or black-hole entropy reviewer can classify as known, false, missing-hypothesis, or useful. It does not modify any frozen Team 1 manuscript, Lean archive, public mirror manifest, or SHA-256 package.

1 Primary sources

The boundary below was checked against Witten's crossed-product black-hole entropy construction [1], Longo–Witten continuous entropy [2], the large- N generalized-entropy algebra of Chandrasekaran–Penington–Witten [3], and the JT-gravity Type II algebra analysis of Penington–Witten [4]. These works are used here as positive constructions for specified algebras and traces, not as claims that finite-center calibration is absent from physics.

2 Finite centers in the crossed product

Let $\mathcal{M}$ be a von Neumann algebra with faithful normal state ω , and let

$$\mathcal{K} = \mathcal{M} \rtimes_{\sigma^\omega} \mathbb{R}$$

be its modular crossed product. For $n \geq 2$, let

$$\mathcal{B}_n = \mathcal{M} \bar{\otimes} \ell^\infty(\{1, \dots, n\}), \quad \Omega_r = \omega \otimes r,$$

where $r = (r_1, \dots, r_n)$ is a faithful probability vector.

Theorem 1 (finite central cores). *There is a canonical crossed-product identification*

$$\mathcal{B}_n \rtimes_{\sigma^{\Omega_r}} \mathbb{R} \cong \mathcal{K} \bar{\otimes} \ell^\infty(\{1, \dots, n\}).$$

Consequently the modular crossed product carries the finite center into the Type II core. It does not by itself select the finite central cardinality, the predicate $n = 2$, or the finite-center trace normalization.

Proof. The modular automorphism group of $\Omega_r = \omega \otimes r$ is

$$\sigma_t^{\Omega_r} = \sigma_t^\omega \otimes \text{id}$$

because the finite abelian factor is central and r is diagonal on it. Crossed products commute with tensoring by a finite algebra on which the action is trivial, giving the displayed isomorphism. The right-hand side still has central projections $1 \otimes e_i$, so the finite center has not been selected or removed by the passage to the core. $\square$

3 Trace calibration fiber

Assume $\mathcal{K}$ is equipped with a faithful normal semifinite trace τ . On the full finite central core, consider traces

$$\tau_a(x_1, \dots, x_n) = \sum_{i=1}^n a_i \tau(x_i), \quad a_i > 0.$$

For a central probability vector $p = (p_i)$, the part of the finite-center entropy that changes with the central trace weights is

$$L_a(p) = \sum_i p_i \log a_i.$$

If the supplied trace calibration consists of central test values

$$T_a(c) = \sum_i a_i c_i, \quad c \in C \subset \mathbb{R}^n,$$

then $L_a(p)$ is determined for all allowed $p \in P$ if and only if

$$T_a|_C = T_b|_C \implies L_a(p) = L_b(p) \quad \text{for every } p \in P. \quad (1)$$

Proposition 1 (diagonal trace tests are insufficient). *Scalar diagonal trace tests do not determine the finite-center entropy normalization. For $n = 2$, the trace weights $a = (1, 3)$ and $b = (2, 2)$ agree on every scalar diagonal test, but for $p = (1/2, 1/2)$,*

$$L_a(p) - L_b(p) = \frac{1}{2} \log \frac{3}{4} \neq 0.$$

Proof. A scalar diagonal test has the form $c = (\lambda, \lambda)$, so

$$T_a(c) = \lambda(a_1 + a_2) = 4\lambda = T_b(c).$$

However

$$L_a(1/2, 1/2) - L_b(1/2, 1/2) = \frac{1}{2}(\log 1 + \log 3 - \log 2 - \log 2) = \frac{1}{2} \log \frac{3}{4}.$$

Thus the same diagonal Type II trace data can leave different absolute finite-center entropy terms. $\square$

Proposition 2 (atomic calibration recovers the finite-center entropy term). *Assume the full finite central core $\mathcal{K} \bar{\otimes} \ell^\infty(\{1, \dots, n\})$ is retained and the supplied calibration contains, for some positive $h \in \mathcal{K}$ with $0 < \tau(h) < \infty$, the atomic trace values*

$$\tau_a(h \otimes e_i) = a_i \tau(h), \quad i = 1, \dots, n.$$

Then the central trace weights a_i are determined up to the same convention used to normalize $\tau(h)$. In particular, for every retained central probability vector p , the finite-center entropy contribution

$$L_a(p) = \sum_i p_i \log a_i$$

is determined. If only the projective class $[a]$ is fixed, then $L_a(p)$ is determined up to a common additive constant independent of p , and all differences $L_a(p) - L_a(p')$ are determined exactly.

Proof. Dividing the supplied value $\tau_a(h \otimes e_i)$ by the fixed normalization $\tau(h)$ gives a_i . Substitution gives $L_a(p)$. If the trace is specified only up to common rescaling $a \mapsto \lambda a$, then

$$L_{\lambda a}(p) = \sum_i p_i \log(\lambda a_i) = L_a(p) + \log \lambda,$$

because $\sum_i p_i = 1$. The shift is independent of p , so it cancels in differences between retained central states. $\square$

4 Entropy-density boundary

Let D be the crossed-product, relative-entropy, or operational data supplied in a comparison class, and let

$$c(X) = \lim_{\Sigma} \frac{S_X(\Sigma)}{\text{Area}(\Sigma)}$$

be an asserted absolute entropy-density coefficient. Then c is determined by D exactly when it is constant on D -fibers:

$$D(X) = D(Y) \implies \lim_{\Sigma} \frac{S_X(\Sigma) - S_Y(\Sigma)}{\text{Area}(\Sigma)} = 0. \quad (2)$$

Finite-center cardinality terms, trace-weight terms such as $L_a - L_b$, and affine entropy or area counterterms are obstructions precisely when their normalized limits in (2) are nonzero.

5 Positive exits and referee question

Exit premise	Effect
Full Type II algebra retained	The finite central summands are part of the specified entropy algebra, rather than omitted before the entropy calculation.
Atomic trace calibration	Tests $h \otimes e_i$ recover the central trace weights a_i , hence the finite-center entropy contribution, by Proposition 2.
Absolute area/entropy calibration	An affinely spanning calibrated state family fixes the additive or affine entropy residual.
Quotient, factor, or inadmissibility axiom	The finite-center comparison class is explicitly removed from the target theory.

The precise external question is:

Does the crossed-product or gravitational Type II construction being invoked in the target setting supply the finite central trace/area calibration above, or does it define entropy only for the algebra and trace normalization already specified?

If the answer is the second, Team 1 has no new crossed-product theorem. Its surviving claim is only the claim-boundary warning: crossed-product entropy should not be overread as finite-center maximality or as an absolute finite-center calibration principle.

References

- [1] E. Witten, *Gravity and the Crossed Product*, JHEP 10 (2022), 008, arXiv:2112.12828.
- [2] R. Longo and E. Witten, *A note on continuous entropy*, arXiv:2202.03357.
- [3] V. Chandrasekaran, G. Penington, and E. Witten, *Large N algebras and generalized entropy*, JHEP 04 (2023), 009, arXiv:2209.10454.
- [4] G. Penington and E. Witten, *Algebras and States in JT Gravity*, arXiv:2301.07257.